

Street and in the skyscrapers on the southern tip of Manhattan could travel down into the city in their cars.

Unfortunately (or perhaps fortunately), however beautiful it was, it was not especially well built, and it was worse maintained. That day in December 1973, as a heavy dump truck overloaded with nine tons of asphalt went over it, several of the supports failed, and the truck plunged onto the street below, followed by a car that was evidently following rather too closely. Sam Schwartz, who visited the site, says it looked as though a new exit ramp had been built. By sheer chance, nobody was killed—a rare miracle in the history of road collapses.

But it did cause an almighty traffic jam, at least on the day itself. The West Side Highway was not a lightly used route. Each day it carried around 70,000 vehicles up and down Manhattan. In a moment of immense, if accidental, foresight, just three months before, Mayor John Lindsay had begged drivers to try to avoid it, as it was so jam-packed. "If you try to use the West Side Highway in the next three months," he warned prophetically, "you'll regret it." Hundreds of vehicles had to reverse their way back off the road, over a period of several hours, with the drivers presumably wondering if their section of the tarmac might be next to falter. The City of New York then immediately closed the road, and for the next couple of decades it sat, derelict. Before it was finally replaced, in the 1990s, it was practically a perfect symbol of the Big Apple's decline.

What it also could become, however, is a symbol of how more roads do not solve congestion. When the road collapsed, one of the immediate fears of New York's traffic engineers was that the 70,000 vehicles it carried each day would reappear elsewhere in Manhattan, clogging up streets with more traffic. But in fact, after that initial jam caused by all of the cars trying to back up off the broken bridge, there were no more. The reality, as Schwartz wrote, was that "the predicted traffic disaster never appeared." The cars "went somewhere, but to this day we have no idea where." Over the next few years, even though the road had disappeared, the number of people traveling into Manhattan's southern business district actually increased. It just happened that fewer of them were doing so in cars.

This reveals something about traffic, and the problem of what is usually called "induced demand." At the margin, whenever they set out on a journey,

people choose between different forms of travel. In a city like New York, where walking, bicycling, or taking the subway or bus are all plausible options for getting somewhere, most people have alternatives to getting in their cars. If you think that there will be a traffic jam, you might choose to take the subway instead. If you know that the road will be clear, you might decide to get into your car instead of taking the subway, or cycling, or whatever.

The result is that the level of congestion on a particular road is essentially priced in. How many lanes a road has does not actually have that much of an impact on how jammed up it gets. Instead, what determines the traffic level is what other options drivers have. Traffic jams stop growing at the point at which the marginal driver decides it would be better to walk, or to hop on a train, or to simply go somewhere else instead. So what happens if you expand a road by adding more lanes? Initially, people carry on with their old travel habits, and so traffic on the road speeds up a lot. But over time—perhaps just a few months—what happens is that people notice that they can get somewhere a lot faster on the new, wider road. Pretty quickly, congestion is just as bad as it was before—because that is the level of congestion needed to deter more people from driving. *

There is relatively reasonable academic evidence for this effect, though the scale and context is often disputed. A study commissioned by the British Department for Transport (an organization which rather likes building roads) tentatively found that a 10 percent increase in road capacity probably induces about a 2 percent increase in the amount of driving, overall—that is, 20 percent of the growth is used up by more cars. That, however, was based on other studies that looked at whole highway networks (in this case, of the United States and the Netherlands), rather than specifically urban roads. It makes sense that induced demand is less likely to be a problem in rural areas, where everyone drives anyway, and cannot switch to taking a subway train instead. In urban areas, other estimates put the figure as high as 40 percent after six months and 100 percent after four years. That is, over four years, the entire extra capacity generated by a road expansion is used up again. Even the Department for Transport study found that "induced demand is likely to be higher for capacity improvements in urban areas or on highly congested routes."

Induced demand is not inherently a bad thing, obviously. If you build a road to, say, a rural village in Tanzania, you would hope that as a result, more